

MA4P Self-Assessment – Jacob Booth 221530207

The module content mostly relied on specific formulae outlined in the key module concepts, this assessment required me to know how to apply them. Since it's all about breaking down larger problems into smaller ones, performing simple multiplication didn't require a calculator for assistance, I just needed the structure of each formula, largely visual problem-solving to begin with. I used the same calculator as usual for a couple of validations before submitting.

8/4/25, 3:18 PM

SIT292 Module 4 Self-Assessment

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Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	1.7/2	
Question 2	1/1	
Question 3	1/1	
Question 4	2/2	
Question 5	1/1	
Question 6	1/1	
Question 7	2/2	
Total	9.7/10	(96%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 4 Self-Assessment
Session ID:	8158182370939113
Exam Start:	Mon Aug 04 2025 14:09:05
Exam Stop:	Mon Aug 04 2025 15:18:33
Time Spent:	1:09:28

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<https://www.newcastle.ac.uk>).

Question 1

Consider the matrix

$$A = \begin{pmatrix} 5 & -3 & -3 \\ -3 & 2 & -3 \\ 5 & -3 & 1 \end{pmatrix}$$

Evaluate the determinant by expanding along row 3.

$$\begin{aligned}
 \det(A) &= \boxed{5} \checkmark \quad \text{Expected answer: } \underline{5} \times \det \\
 &\begin{pmatrix} \boxed{-3} & \boxed{-3} \\ \boxed{2} & \boxed{-3} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \underline{-3} & \underline{-3} \\ \underline{2} & \underline{-3} \end{pmatrix} + \\
 &\boxed{-3} \times \quad \text{Expected answer: } \underline{3} \times \det \\
 &\begin{pmatrix} \boxed{5} & \boxed{-3} \\ \boxed{-3} & \boxed{-3} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \underline{5} & \underline{-3} \\ \underline{-3} & \underline{-3} \end{pmatrix} + \\
 &\boxed{1} \checkmark \quad \text{Expected answer: } \underline{1} \times \det \\
 &\begin{pmatrix} \boxed{5} & \boxed{-3} \\ \boxed{-3} & \boxed{2} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \underline{5} & \underline{-3} \\ \underline{-3} & \underline{2} \end{pmatrix} \\
 &= \boxed{5} \checkmark \quad \text{Expected answer: } \underline{5} \times \\
 &\boxed{15} \checkmark \quad \text{Expected answer: } \underline{15} + \\
 &\boxed{-3} \times \quad \text{Expected answer: } \underline{3} \times \boxed{-24} \checkmark \quad \text{Expected answer: } \underline{-24} \\
 &+ \boxed{1} \checkmark \quad \text{Expected answer: } \underline{1} \times \\
 &\boxed{1} \checkmark \quad \text{Expected answer: } \underline{1} \\
 &= \boxed{4} \checkmark \quad \text{Expected answer: } \underline{4}
 \end{aligned}$$

Note: If the coefficient of any sub-matrix determinant is zero, you will still need to include the submatrix in the first line of working and include its determinant in the second line of working.

Score: 1.7/2 ✓**Gap 0**

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 2

- ✗ Your answer is incorrect.

Gap 3

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 4

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 5

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 6

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 7

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 8

- ✗ Your answer is incorrect.

Gap 9

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 10

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 11

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 12

✓ Your answer is correct. You were awarded **0.2** marks.

You scored **1.7** marks for this part.

Question 2

Consider the following matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

If $\det(A) = -8$, use Theorem 3.1.2 to determine the value of

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{11} & a_{14} \\ a_{21} & a_{22} & a_{21} & a_{24} \\ a_{31} & a_{32} & a_{31} & a_{34} \\ a_{41} & a_{42} & a_{41} & a_{44} \end{pmatrix}$$

Which property can be used?

- ☐ 1. Row/Column of Zeros
- ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant
- ☒ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column



Expected answer:

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant
- ☒ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column

What is the value of the resulting determinant?

0

Expected answer: 0

Score: 1/1

Gap 0

- You chose a correct answer. You were awarded **0.3** marks.

Gap 1

- Your answer is correct. You were awarded **0.7** marks.

You scored **1** mark for this part.

Question 3

Consider the following matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

If $\det(A) = -15$, use Theorem 3.1.2 to determine the value of

$$\det \begin{pmatrix} a_{11} & a_{12} & -5a_{13} & a_{14} \\ a_{21} & a_{22} & -5a_{23} & a_{24} \\ a_{31} & a_{32} & -5a_{33} & a_{34} \\ a_{41} & a_{42} & -5a_{43} & a_{44} \end{pmatrix}$$

Which property can be used?

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☒ 3. Row/Column multiplied by a constant
- ☐ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column



Expected answer:

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☒ 3. Row/Column multiplied by a constant
- ☐ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column

What is the value of the resulting determinant?

75



Expected answer: 75

Score: 1/1 ✓

Gap 0

- ✓ You chose a correct answer. You were awarded **0.3** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.7** marks.

You scored **1** mark for this part.

Question 4

For the following matrix,

$$A = \begin{pmatrix} 9 & 8 & 9 \\ 3 & 3 & -3 \\ 7 & 5 & -7 \end{pmatrix}$$

First, calculate the cofactors of A ,

$c_{11} =$ -6 ✓ Expected answer: -6	$c_{12} =$ 0 ✓ Expected answer: 0	$c_{13} =$ -6 ✓ Expected answer: -6
$c_{21} =$ 101 ✓ Expected answer: 101	$c_{22} =$ -126 ✓ Expected answer: -126	$c_{23} =$ 11 ✓ Expected answer: 11
$c_{31} =$ -51 ✓ Expected answer: -51	$c_{32} =$ 54 ✓ Expected answer: 54	$c_{33} =$ 3 ✓ Expected answer: 3

Hence give the adjugate of A ,

$\text{adj}A =$

Rows: Columns:



Expected answer:

-6	101	-51
0	-126	54
-6	11	3

Score: 2/2 ✓

Gap 0

✓ Your answer is correct. You were awarded 0.15 marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 2

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 3

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 4

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 5

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 6

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 7

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 8

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 9

- ✓ Your answer is correct. You were awarded **0.65** marks.

You scored **2** marks for this part.

Question 5

Use determinants to find a value of a for which the matrix is **not** invertible.

$$\begin{pmatrix} 1 & -3 & 7 \\ 3 & -3 & 6 \\ 5 & 9 & a \end{pmatrix}$$

Enter the your value of a exactly, using fractions if required.

$a =$ ✓

Expected answer:

Score: 1/1 ✓

✓ Your answer is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Question 6

Let $A, B \in \mathbf{M}_{3,3}(\mathbb{R})$. Suppose that $\det(A) = 4$ and $\det(B) = -2$.

i) $\det(AB) =$ ✓

Expected answer:

ii) $\det(2A) =$ ✓

Expected answer:

iii) $\det(A^T B^{-1}) =$ ✓

Expected answer:

Score: 1/1 ✓

Gap 0

✓ Your answer is correct. You were awarded 0.3 marks.

Gap 1

✓ Your answer is correct. You were awarded 0.35 marks.

Gap 2

✓ Your answer is correct. You were awarded 0.35 marks.

You scored 1 marks for this part.

Question 7

Apply the LU-Algorithm to factorise the following matrix.

$$A = \begin{pmatrix} 3 & -3 & -6 & 9 \\ 3 & -4 & -8 & 6 \\ -2 & 3 & 6 & -3 \end{pmatrix}$$

Give the matrices L and U that result.

$L =$

Rows: Columns:

$$\begin{pmatrix} 3 & 0 & 0 \\ 3 & -1 & 0 \\ -2 & 1 & 1 \end{pmatrix} \quad \checkmark$$

Expected answer:

$$\begin{pmatrix} 3 & 0 & 0 \\ 3 & -1 & 0 \\ -2 & 1 & 1 \end{pmatrix}$$

U =

Rows: 3 Columns: 4

1	-1	-2	3
0	1	2	3
0	0	0	0



Expected answer:

1	-1	-2	3
0	1	2	3
0	0	0	0

Score: 2/2 ✓

Gap 0

✓ Your answer is correct. You were awarded 1 mark.

Gap 1

✓ Your answer is correct. You were awarded 1 mark.

You scored 2 marks for this part.

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Module 4 Assessment

$$1) A = \begin{pmatrix} 5 & -3 & -3 \\ -3 & 2 & -3 \\ 5 & -3 & 1 \end{pmatrix}$$

$$a_{31} = 5$$

$$a_{32} = -3$$

$$a_{33} = 1$$

$$M_{31} = \begin{pmatrix} -3 & -3 \\ 2 & -3 \end{pmatrix}$$

$$M_{32} = \begin{pmatrix} 5 & -3 \\ -3 & -3 \end{pmatrix}$$

$$\det(M_{31}) = (-3)(-3) - (-3)(2) = 9 + 6 = 15$$

$$\det(M_{32}) = (5)(-3) - (-3)(-3) = -15 - 9 = -24$$

$$\det(M_{33}) = (5)(2) - (-3)(-3) = 10 - 9 = 1$$

$$a_{31} \times \det(M_{31})$$

$$- a_{32} \times \det(M_{32})$$

$$+ a_{33} \times \det(M_{33})$$

$$M_{33} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}$$

$$\det(A) = 5 \times 15 - (-3) \times (-24) + 1 \times 1 = 75 - 72 + 1 = 4$$

$$A_{11} = \begin{pmatrix} 9 & 8 \\ 7 & 9 \end{pmatrix}, \det A_{11} = (9)(9) - (8)(7) = 81 - 56 = 25 \quad C_{11} = (1)^{1+1} (25) = 25$$

$$A_{12} = \begin{pmatrix} 9 & 9 \\ 3 & -3 \end{pmatrix}, \det A_{12} = (9)(-3) - (9)(3) = -27 - 27 = -54 \quad C_{12} = (1)^{1+2} (-54) = -54$$

$$A_{13} = \begin{pmatrix} 9 & 9 \\ 3 & -3 \end{pmatrix}, \det A_{13} = (9)(-3) - (9)(3) = -27 - 27 = -54 \quad C_{13} = (1)^{1+3} (-54) = -54$$

$$A_{21} = \begin{pmatrix} 9 & 8 \\ 3 & 3 \end{pmatrix}, \det A_{21} = (9)(3) - (8)(3) = 27 - 24 = 3 \quad C_{21} = (1)^{2+1} (3) = 3$$

$$\text{adj } A = C^T$$

$$C = \begin{pmatrix} -6 & 0 & -6 \\ 101 & -126 & 11 \\ -51 & 54 & 3 \end{pmatrix}$$

$$\therefore \text{adj } A = \begin{pmatrix} -6 & 101 & -51 \\ 0 & -126 & 54 \\ -6 & 11 & 3 \end{pmatrix}$$

$$\begin{aligned}
 5) \quad \begin{pmatrix} 1 & -3 & 7 \\ 2 & -3 & 6 \\ 0 & a & a \end{pmatrix} \quad \det &= 1 \cdot ((-3 \times a) - (6 \times 9)) - (-3) \cdot ((3 \times a) - (6 \times 5)) + 7 \cdot ((3 \times 9) - (-3 \times 5)) \\
 &= -3a - 54 - 90 \\
 &= -3a - 144 \\
 &= -3a - 144 = 0 \\
 &\Rightarrow -3a = 144 \\
 &\Rightarrow a = -48
 \end{aligned}$$

$$6) \det(A) = 4, \det(B) = -2$$

$$i) \det(AB) = 4 \times (-2) = -8$$

$$ii) \det(2A) = 2^3 \det(A) = 8 \times 4 = 32$$

$$\begin{aligned}
 iii) \det(A^T B^{-1}) & \quad \det(A^T) = \det A = 4 \\
 & \quad \det(B^{-1}) = \frac{1}{\det B} = \frac{1}{-2} \\
 & \quad \det(A^T) \times \det(B^{-1}) = -2
 \end{aligned}$$

$$\begin{aligned}
 A &= \begin{pmatrix} 3 & -3 & -6 & 9 \\ 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 &\xrightarrow{-(3)ur_1} \begin{pmatrix} 3 & -3 & -6 & 9 \\ 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 &\xrightarrow{-(2)ur_1} \begin{pmatrix} 3 & -3 & -6 & 9 \\ 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 &\xrightarrow{-(1)ur_2} \begin{pmatrix} 3 & -3 & -6 & 9 \\ 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow \begin{pmatrix} 3 & -3 & -6 & 9 \\ 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 &\xrightarrow{/3=ur_1} \begin{pmatrix} 1 & -1 & -2 & 3 \\ 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 &\xrightarrow{/(-1)=ur_2} \begin{pmatrix} 1 & -1 & -2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 &\xrightarrow{-(1)ur_3} \begin{pmatrix} 1 & -1 & -2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

LxU yields A.

$$L = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 1 & -1 & -2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$